

Juan David Muñoz-Bolaños

Rechenauerstraße 42, Innsbruck, Austria (Tirol)

+43 676 9115145

juan.munoz@i-med.ac.at

TriLite Technologies GmbH

Attn: HR / Recruiting Team

Frankenberggasse 13

1040 Vienna, Austria

Subject: Application for Senior Optical Engineer (Vienna)

Dear Hiring Team,

I am writing to express my enthusiasm for the **Senior Optical Engineer** position at TriLite Technologies. As an R&D Optical Engineer specializing in laser beam shaping and MEMS-based optical systems, I have followed TriLite's development of the world's smallest projection displays with great interest. Your mission to unlock the Augmented Reality revolution aligns perfectly with my background in high-precision laser optics and prototyping.

The position requires a candidate who can drive the "design and realization of prototypes" while managing complex optical simulations. My career at the intersection of industrial vision and advanced microscopy prepares me to contribute immediately:

1. Laser Beam Manipulation & MEMS Integration:

TriLite's technology relies on precise laser beam scanning. In my current PhD research, I design complex optical setups utilizing **Adaptive Optics** and **SLM/DMD (MEMS)** technology. I currently integrate high-speed Texas Instruments MEMS devices to correct optical aberrations and shape laser wavefronts. This hands-on experience with beam steering and shaping is directly transferable to the LBS (Laser Beam Scanning) challenges in AR display systems.

2. Optical Simulation & Tolerance Analysis:

I bridge the gap between theoretical modeling and physical realization. I have extensive experience simulating complex optical paths (including shaped simulations and phase analysis) to predict system performance before prototyping. I am eager to apply this expertise to perform the tolerance analysis and Zemax simulations required for your projection modules, ensuring alignment with your rigorous performance targets.

3. Industry Experience & Agility:

Prior to my research, I worked as a Machine Vision Engineer at INTECOL S.A.S., where I delivered robust **MVTec HALCON** solutions for high-speed industrial lines. This industrial background ensures I understand the difference between a "lab experiment" and a "product-ready" system. I thrive in dynamic, cross-functional environments and look forward to the "startup-like culture" at TriLite, collaborating with electronics and mechanical teams to push the technology envelope.

4. Local Integration:

I am currently based in Austria and, with a MINT degree from Germany and an Austrian PhD, I qualify for the **Red-White-Red Card (High-Skilled)**. This ensures a straightforward hiring process. I am eager to relocate to Vienna to join your international team and contribute to the next generation of display technology.

I would be welcome the opportunity to discuss how my experience in precision laser optics can support TriLite's growth ambitions. Thank you for considering my application.

Sincerely,

Juan David Muñoz-Bolaños
R&D Optical Engineer
Medical University of Innsbruck